

BDD

Création et manipulation des vues

```
mysql> INSERT INTO V_General_Average VALUES('nermine', 'kirouane', 15);
ERROR 1471 (HY000): The target table V_General_Average of the INSERT is
not insertable-into
mysql> select * from V_Student_List;
+-----+-----+
| student_name | student_first_name |
+-----+-----+
| BOUSSAI      | MOHAMED             |
| CHAID        | LAMIA               |
| BRAHIMI      | SOUAD               |
| LAMA         | SAID                |
| KHELIFI      | NORA                |
| BENALI       | YACINE              |
+-----+-----+
6 rows in set (0.00 sec)

mysql> DROP view V_Student_List;
Query OK, 0 rows affected (0.01 sec)

mysql> DROP view V_Average;
Query OK, 0 rows affected (0.01 sec)
```

KIROUANE NERMINE

2026/04/23

Création et manipulation des vues

Tâche 0 : Gestion des utilisateurs et base de données

1. Exécutez le script

```
mysql> tee C:/Users/HP/OneDrive/Documents/BDD/spool6.txt
Logging to file 'C:/Users/HP/OneDrive/Documents/BDD/spool6.txt'
mysql> source C:/Users/HP/OneDrive/Documents/BDD/TP5_script.sql
Query OK, 1 row affected (0.09 sec)

Database changed
Query OK, 0 rows affected (0.12 sec)

Query OK, 0 rows affected (0.04 sec)

Query OK, 0 rows affected (0.08 sec)

Query OK, 0 rows affected (0.08 sec)

Query OK, 6 rows affected (0.03 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

executing using command: SOURCE

saving the text in .txt file using the command : TEE

2. Créer un utilisateur nommé "BDDAdmin" avec le mot de passe "admin1234".

```
mysql> create user 'BDDAdmin' identified by 'admin1234';
Query OK, 0 rows affected (0.07 sec)

mysql> |
```

i created the user using the commands:

create user (name) identified by (password)

```
mysql> select user from mysql.user;
```

user
BDDAdmin
TP3user1
dounia
nermine
mysql.infoschema
mysql.session
mysql.sys
root

```
8 rows in set (0.01 sec)
```

3. Accordez à l'utilisateur "BDDAdmin" l'ensemble des privilèges sur la base de données.

```
mysql> show databases;
```

Database
cinema
execution_db
information_schema
mysql
performance_schema
student_db
tp3
uni2
university_db
usthb_cs_bd
usthb_info_db

```
11 rows in set (0.04 sec)

mysql>
```

we have the database (student_db)

we connect

```
mysql> use student_db;
Database changed
mysql> grant all privileges on student_db.* TO 'BDDAdmin';
Query OK, 0 rows affected (0.01 sec)

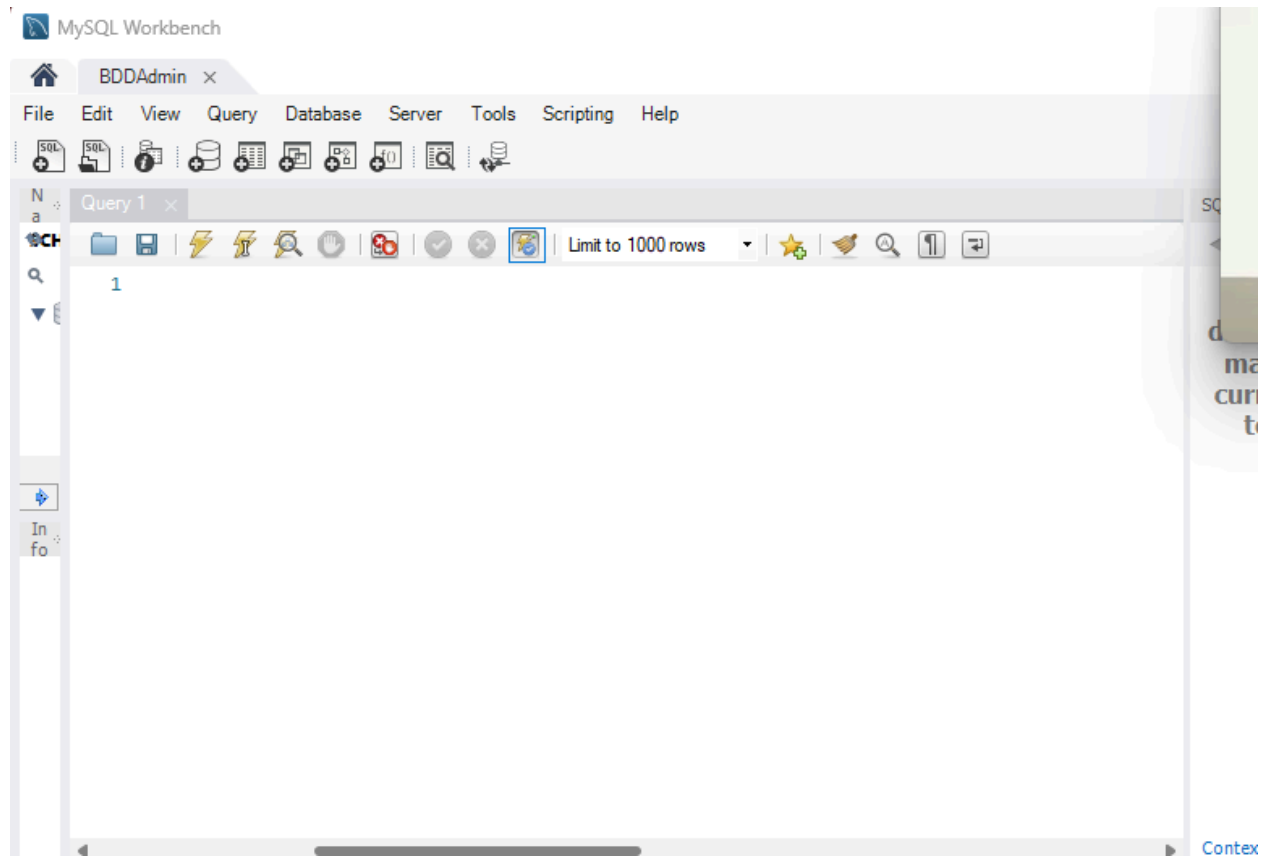
mysql>
```

and i gave all privileges the user (student_db)

4. Affichez la liste des bases de données et connectez vous à la base d'étudiants en tant qu'utilisateur "BDDAdmin".

```
mysql> show databases;
+-----+
| Database |
+-----+
| cinema   |
| execution_db |
| information_schema |
| mysql    |
| performance_schema |
| student_db |
| tp3      |
| uni2     |
| university_db |
| usthb_cs_bd |
| usthb_info_db |
+-----+
11 rows in set (0.00 sec)

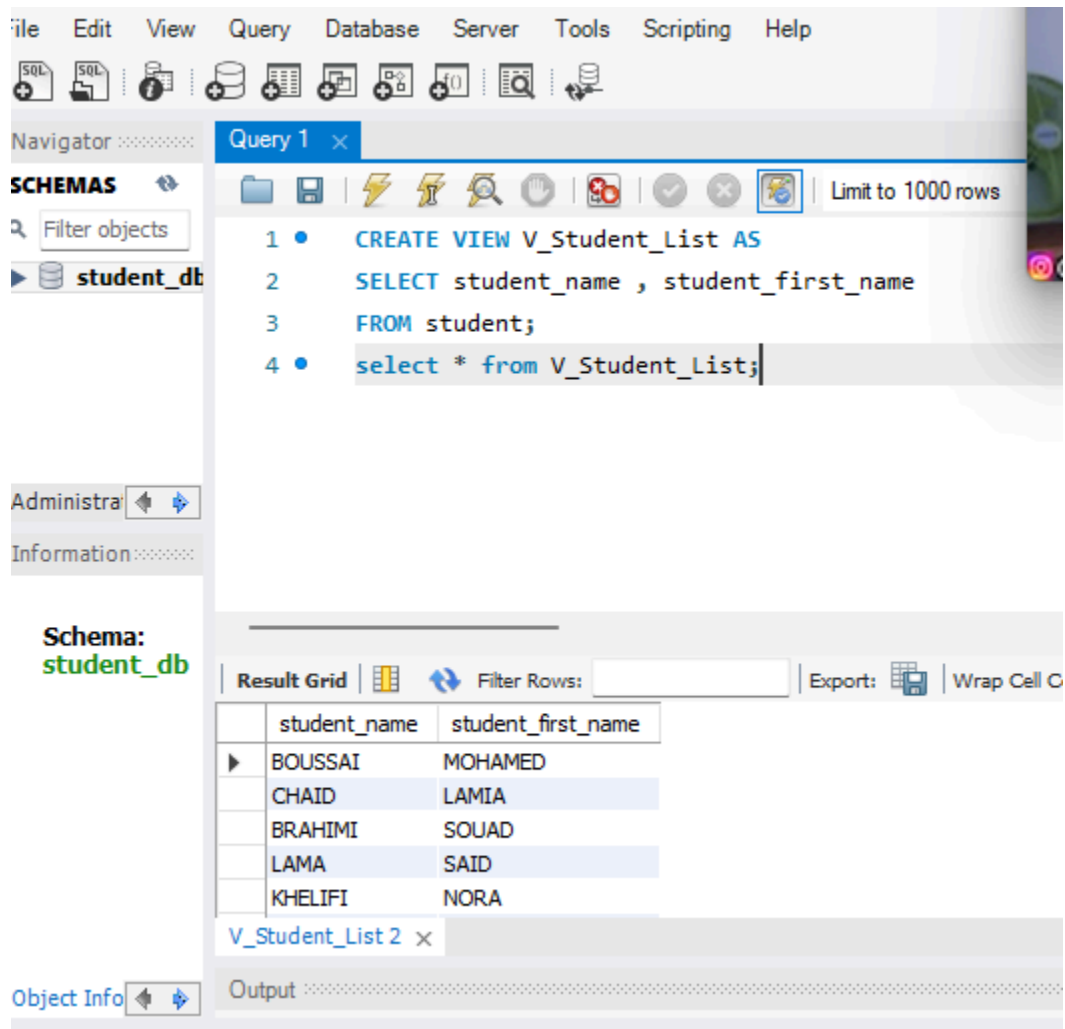
mysql>
```



connected to the user BDDAdmin.

Tâche 1 : Création des Vues ;

1. Créez la vue V_Student_List qui contient les noms et les prénoms de tous les étudiants.



i created the view using the teacher syntax

```
-- Syntax:
CREATE VIEW view_name AS
SELECT column_list FROM table_name WHERE condition(s);
```

and i specified and stored first name and last name

and we can see that the view is created

4 • `select * from V_Student_List;`

Result Grid | Filter Rows: | Export:

	student_name	student_first_name
▶	BOUSSAI	MOHAMED
	CHAID	LAMIA
	BRAHIMI	SOUAD
	LAMA	SAID
	KHELIFI	NORA

V Student List 2 x

2. Créez la vue V_Selected_Modules qui contient les informations des modules : POO, BDD ou RESEAU.

```
19990011 | FATHI | OMAR | 50 |
20000001 | HAROUNI | AMINE | 45 |
20160022 | MANSOURI | KARIM | 48 |
20170015 | ARABI | ZOUBIDA | 39 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> select * from unit;
+-----+-----+-----+-----+
| unit_code | module | nbr_hours | teacher_ID |
+-----+-----+-----+-----+
| FEI0001 | POO | 6 | 20000001 |
| FEI0002 | BDD | 6 | 19990011 |
| FEI0003 | RESEAU | 3 | 20170015 |
| FEI0004 | SYSTEME | 6 | 19980078 |
| FEI0005 | SECURITY | 4 | 20160022 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

after searching i found that module informations are stored at unit:

```
5
6 • CREATE VIEW V_Selected_Modules AS
7 SELECT *
8 FROM unit
9 WHERE module IN ('POO', 'BDD', 'RESEAU');
```

so we have a small version of the table unit that only have poo , bdd , reseau

```
mysql> select * from V_Selected_Modules;
+-----+-----+-----+-----+
| unit_code | module | nbr_hours | teacher_ID |
+-----+-----+-----+-----+
| FEI0001   | POO    | 6         | 20000001   |
| FEI0002   | BDD    | 6         | 19990011   |
| FEI0003   | RESEAU | 3         | 20170015   |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

the view is created succefully

3. Créez la vue V_Average qui contient pour chaque étudiant : le nom, le prénom, le titre du cours, ainsi que la moyenne calculée par module. Query SQL à utiliser :

```
SELECT student_name, student_first_name, module,
       (CC_mark + PW_mark + (exam_mark * 2)) / 4 AS Average
FROM student, studentUnit, unit
WHERE student.student_ID = studentUnit.student_ID
AND studentUnit.unit_code = unit.unit_code;
```



```

11  SELECT
12      student_name,
13      student_first_name ,
14      module,
15      (CC_mark + PW_mark + exam_mark * 2) / 4 AS Average
16  FROM student , studentunit, unit
17  WHERE student.student_ID = studentunit.student_ID
18  AND studentunit.unit_code = unit.unit_code;
19  • SELECT * from V_Average;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [FA](#)

	student_name	student_first_name	module	Average
▶	BOUSSAI	MOHAMED	POO	10.7500
	CHAIID	LAMIA	POO	13.2500
	LAMA	SAID	POO	15.5000
	KHELIFI	NORA	POO	13.7500
	BOUSSAI	MOHAMED	BDD	12.0000

V_Average 3 x

Output:

we only named the view V_Average and used the teacher structure to create a view that hold the average moy for each student

4. Créez la vue V_General_Average qui contient les noms, les prénoms des étudiants, ainsi que leur moyenne générale. Query SQL à utiliser

```

SELECT student_name, student_first_name, AVG(Average) AS Gen_Avg
FROM V_Average
GROUP BY student_id, student_name, student_first_name;

```

- ```

CREATE VIEW V_General_Average AS
SELECT
 student_name,
 student_first_name,
 AVG(Average) AS GEN_AVG
FROM V_Average
GROUP BY student_name, student_first_name;

```

we created the view the teacher told us about without student\_id

## Tâche 2 : Consultation et Manipulation des vues

1. Affichez le contenu de la vue V\_Student\_List.

```
mysql> select * from V_Student_List;
+-----+-----+
| student_name | student_first_name |
+-----+-----+
BOUSSAI	MOHAMED
CHAID	LAMIA
BRAHIMI	SOUAD
LAMA	SAID
KHELIFI	NORA
BENALI	YACINE
+-----+-----+
6 rows in set (0.00 sec)

mysql> |
```

view created successfully

2. Affichez le contenu de la vue V\_Average.

```
mysql> select * from V_Average;
```

| student_name | student_first_name | module   | Average |
|--------------|--------------------|----------|---------|
| BOUSSAI      | MOHAMED            | P00      | 10.7500 |
| CHAID        | LAMIA              | P00      | 13.2500 |
| LAMA         | SAID               | P00      | 15.5000 |
| KHELIFI      | NORA               | P00      | 13.7500 |
| BOUSSAI      | MOHAMED            | BDD      | 12.0000 |
| CHAID        | LAMIA              | BDD      | 15.0000 |
| BRAHIMI      | SOUAD              | BDD      | 16.0000 |
| BRAHIMI      | SOUAD              | RESEAU   | 11.7500 |
| BENALI       | YACINE             | RESEAU   | 10.0000 |
| CHAID        | LAMIA              | SYSTEME  | 14.5000 |
| BRAHIMI      | SOUAD              | SYSTEME  | 14.7500 |
| LAMA         | SAID               | SYSTEME  | 16.2500 |
| KHELIFI      | NORA               | SECURITY | 16.5000 |
| BENALI       | YACINE             | SECURITY | 13.7500 |

```
14 rows in set (0.00 sec)
```

```
mysql> |
```

3. Affichez le contenu de la vue V\_General\_Average.

```
mysql> select * from V_General_Average;
```

| student_name | student_first_name | GEN_AVG     |
|--------------|--------------------|-------------|
| CHAID        | LAMIA              | 14.25000000 |
| BRAHIMI      | SOUAD              | 14.16666667 |
| LAMA         | SAID               | 15.87500000 |
| BOUSSAI      | MOHAMED            | 11.37500000 |
| KHELIFI      | NORA               | 15.12500000 |
| BENALI       | YACINE             | 11.87500000 |

```
6 rows in set (0.00 sec)
```

4. Affichez le contenu de la vue V\_Selected\_Modules.

```
mysql> select * from V_Selected_Modules;
```

| unit_code | module | nbr_hours | teacher_ID |
|-----------|--------|-----------|------------|
| FEI0001   | POO    | 6         | 20000001   |
| FEI0002   | BDD    | 6         | 19990011   |
| FEI0003   | RESEAU | 3         | 20170015   |

```
3 rows in set (0.00 sec)
```

```
mysql>
```

5. Tentez d'insérer une nouvelle ligne dans la vue V\_Student\_List. — Question :  
L'insertion est-elle acceptée? Pourquoi?

```
mysql> describe V_Student_List;
```

| Field              | Type        | Null | Key | Default | Extra |
|--------------------|-------------|------|-----|---------|-------|
| student_name       | varchar(50) | YES  |     | NULL    |       |
| student_first_name | varchar(50) | YES  |     | NULL    |       |

```
2 rows in set (0.01 sec)
```

```
mysql>
```

```
mysql> INSERT INTO V_Student_List Values('nermine', 'kirouane');
```

```
ERROR 1423 (HY000): Field of view 'student_db.v_student_list' underlying
table doesn't have a default value
```

```
mysql> |
```

the insert failed

so insert is not accepted to view

because view do not include all columns and inserting into table will fail so if wanna insert we need to insert into the original table

6. Tentez d'insérer une ligne dans la vue V\_Average. — Question : Quel est le problème rencontré lors de cette insertion?

```
14 rows in set (0.00 sec)

mysql> INSERT INTO V_Average VALUES ('nermine', 'kirouane', 'P00', 17.7)
;
ERROR 1394 (HY000): Can not insert into join view 'student_db.v_average'
without fields list
mysql>
```

we cant insert into V\_Average

view is based on many tables so mysql cannot insert into multiples tables

we cant insert into the calculated column

```
(CC_mark + PW_mark + (exam_mark * 2)) / 4 AS Average
```

that why insert failed

so the insert failed because:

- view contain join

- view contain calculated column

- we cant update view

7. Tentez d'insérer une ligne dans la vue V\_General\_Average. — Question : Pourquoi le système refuse-t-il la modification de données via cette vue?

```
6 rows in set (0.00 sec)

mysql> INSERT INTO V_General_Average VALUES('nermine', 'kirouane', 15);
ERROR 1471 (HY000): The target table V_General_Average of the INSERT is
not insertable-into
mysql> |
```

```
select * from V_General_Average;
INSERT INTO V_General_Average VALUES('nermine', 'kirouane', 15);
```

the problem is:

we cant insert into view

we cant insert into view that contain fonction d'agrégation

we cant insert into table that contain GROUP BY

the view depend on another view V\_Average

8. Supprimez toutes les vues créées précédemment

```
29 • INSERT INTO V_General_Average VALUES('nermine', 'kirouane', 15);
30 • DROP view V_Student_List;
31 • DROP view V_Average;
32 • DROP view V_General_Average;
33 • DROP view V_Selected_Modules;
34
```

\$

i used the teacher commands here

## 6. DELETING A VIEW

When a view is no longer needed, you can remove it using the `DROP VIEW` statement.

```
DROP VIEW view_name;
```

**Important Note :** Dropping a view **does not** delete or affect the actual data in your database. It only deletes the named relational structure.

```
mysql> DROP view V_Student_List;
Query OK, 0 rows affected (0.01 sec)

mysql> DROP view V_Average;
Query OK, 0 rows affected (0.01 sec)

mysql> DROP view V_Selected_Modules;
Query OK, 0 rows affected (0.01 sec)

mysql> DROP view V_General_Average;
ERROR 1051 (42S02): Unknown table 'student_db.v_general_average'
mysql> DROP view V_General_Average;
ERROR 1051 (42S02): Unknown table 'student_db.v_general_average'
mysql> DROP view V_General_Average;
ERROR 1051 (42S02): Unknown table 'student_db.v_general_average'
mysql> select * from V_Student_List;
ERROR 1146 (42S02): Table 'student_db.v_student_list' doesn't exist
mysql>
```

view do not exist after the drop.